

財政學 HW6

1. 試舉一例說明利率小於等於內部報酬率，但淨現值為負的情形 [提示：比較情境 A 與情境 B。情境 A：第零期只有成本，第一期只有利益；情境 B：第零期只有利益，第一期只有成本]
2. Conduct a project evaluation with all three cost-benefit analysis methods using the information provided in the last two slides (Hint: 用 Excel 計算上課投影片最後兩頁的資訊)
3. (Rosen 8.3) A project yields an annual benefit of \$25 a year, starting next year and continuing forever. What is the present value of the benefits if the interest rate is 10 percent? [Hint: 由等比級數公式推導 The infinite sum $x+x^2+x^3+\dots$ is equal to $x/(1-x)$, where x is a number less than 1.] Generalize your answer to show that if the perpetual annual benefit is B and the interest rate is r , then the present value is B/r .
4. (Rosen 8.4) Suppose that you are planning to take a year vacation to bike across the United States. Someone is willing to sell you a new bicycle for \$500. At the end of the year, you expect to resell the bicycle for \$350. The benefit to you of using the bicycle is the equivalent of \$170.
 - (a) What is the internal rate of return?
 - (b) If the discount rate is 5 percent, should you buy the bicycle?
5. (Hyman 6.2) Suppose a proposed new road to be constructed in North Carolina between Raleigh and Morehead City will lower the average cost per trip by car from \$5 to \$4. Currently, 500,000 trips are made between the two cities per year. An estimate indicates that, all other things being equal, the new road will increase the number of trips per year to 600,000. Calculate the annual benefits to motorists of the new road as based on their willingness to pay.